AB-220 analytical balance user manual

Fictional document for Northfield Lab Supply, a fictional company created for a portfolio project. Not for real use.

Item	Value
Document ID	NLS-DOC-AB220-MAN-R3
Product	Analytical Balance AB-220
Applies to	NLS-AB220-STD, NLS-AB220-CRT
Revision	R3
Effective date	2026-03-01
Status	current

1. Introduction

The AB-220 is an analytical balance with a capacity of 220 g and a readability of 0.1 mg. It is sold in two versions: the Standard and the Certified, which is supplied with a calibration certificate and a longer warranty.

Differences between the versions:

	Standard	Certified
Calibration certificate	Not included	Included (issued with the unit)
Warranty	1 year	2 years
Availability	In stock	Made to order (about 3 working days)

2. Safety

- Use only the mains adaptor supplied. Do not use the balance with a damaged adaptor or cable.
- Operate only within 15 to 30 °C and 20 to 80 % humidity, non-condensing.
- Do not exceed the 220 g capacity; overloading can damage the sensor.
- Keep liquids away from the pan and the casing.
- Disconnect the adaptor before cleaning.

3. Setting up

1. Remove the packaging and check the contents against the packing list.
2. Place the balance on a firm, level bench away from draughts and vibration.
3. Level the balance using the feet until the level indicator is centred.
4. Fit the pan and the draft shield, then connect the adaptor (UK plug).
5. Leave the balance powered for 30 minutes before use.

4. Operation

1. Close the draft shield doors and press Zero.
2. Place the sample or container on the centre of the pan, then close the doors.
3. Wait for the stability indicator and read the value.
4. To calibrate: after warm-up, hold the calibration key and place the 200 g test weight (not supplied) when prompted.
5. Remove the sample and press Zero before the next weighing.

5. Cleaning and maintenance

- Wipe the pan and shield with a lint-free cloth and a mild detergent; dry before use.
- Clean spills immediately. Do not use solvents on the casing.
- Recalibrate after moving the balance or after a large temperature change.

6. Troubleshooting

Problem	Likely cause	What to do
Unstable reading	Draughts or vibration	Close the doors, move away from air outlets and check the bench
Reading drifts	Warm-up too short, or temperature difference with the sample	Allow 30 minutes warm-up; let the sample reach room temperature
Overload shown	Weight above 220 g	Remove the load
Does not switch on	Adaptor not connected	Check the adaptor and socket

7. Warranty and contact

Warranty: Standard: 1 year. Certified: 2 years.

Warranty claims: contact Customer service with the order number and a description of the fault. The warranty covers manufacturing defects and does not cover damage from misuse, spills or unauthorised repair.

Contact: customerservice@northfieldlabsupply.example